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Experiments and Sampling

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**Final Project:**

# **Goal & Age Differential in European Football**

## INTRODUCTION

Optimization problems rely on effective measurements of performance and characteristics of a population to understand the structure of the system that is being studied and its margin of freedom interpreted as the changes that can be done to the system to improve the measured performance. As such, the design of a metric that reflects the nuances of a population as well as the objective of the designer is crucial in the process.

As part of the gamut of disciplines where diagnostic and measurements of performance plays a crucial role, the sports industry stands out as a user and producer of statistics to measure aspects of the games and the players that can be hard to quantify but that, nevertheless, can be approximated. Within this industry, soccer plays a special role as the most popular sport with a following of over 53% of the population worldwide<sup>1</sup> and international competitions that date back almost a century, when the first World Cup was played in 1930. Ever since, the game and the metrics with which performance is measured have changed and are still developing to better understand the underlying variables between measured categories.

Some of the nuances when trying to measure the performance of a team or a player are the different positions of players in the field. Regularly, a soccer match lasts a standard of 90 min, divided in two halves; there are 11 players per team, for a total of 22 players in the field and the typical positions are: Goalkeeper, Right Fullback, Left Fullback, Center back, Center Midfield, Right Midfield/Wing, Forward and Left

Midfield/Wing. Some of these positions can be eliminated or altered, depending on the strategy and formation of a team. As such, to evaluate the performance of a team an analyst can use different outputs such as wins, goals, goals per game, or introduce more factors (touches, interceptions, yellow cards, among others) with specific loadings to create a customized metric.

## **OBJECTIVE**

To choose samples of a size that properly reflect the characteristics of each league and that allows us to evaluate the characteristics of different soccer leagues in Europe by contrasting measurements of performance for the respective leagues in the season 2022-2023.

## **DATA SET**

Observations were compiled from the website: [fbref.com](https://fbref.com)<sup>2</sup>; a site with statistics of each player's performance, each team, and each soccer league around the world. Each observation represents a player, and the number of observations per-data set will depend on the league but will contain at least 200 observations. The variables used to measure performance as well as the variables used to identify the observations: Name of the Player, position, Age, Matches Played, Total Number of Minute Played, Number of Goals, Number of Assists, Goals by Penalty Kick, Number of Penalty Kicks

Attempted Number of Yellow Cards, Number of Red Cards. A glimpse of the Data Set including 10 players is presented below.

| Player                           | Pos   | Age | MP | Min   | Goals | Assists | Goals-Kept | Penalty Kept | Penalty Attempts | Cards Yellow | Cards Red |
|----------------------------------|-------|-----|----|-------|-------|---------|------------|--------------|------------------|--------------|-----------|
| <a href="#">Domingos Duarte</a>  | DF    | 27  | 31 | 2,657 | 0     | 2       | 0          | 0            | 0                | 11           | 1         |
| <a href="#">Takefusa Kubo</a>    | FW,MF | 21  | 35 | 2,439 | 9     | 4       | 9          | 0            | 0                | 3            | 0         |
| <a href="#">Edinson Cavani</a>   | FW    | 35  | 25 | 1,449 | 5     | 2       | 3          | 2            | 3                | 6            | 0         |
| <a href="#">Iglesias</a>         | DF,MF | 24  | 32 | 2,146 | 1     | 0       | 1          | 0            | 0                | 7            | 0         |
| <a href="#">Brian Ocampo</a>     | MF,FW | 23  | 18 | 1,119 | 1     | 1       | 1          | 0            | 0                | 2            | 0         |
| <a href="#">Copete</a>           | DF    | 22  | 28 | 2,206 | 1     | 0       | 1          | 0            | 0                | 7            | 1         |
| <a href="#">Robin Le Normand</a> | DF    | 25  | 33 | 2,796 | 0     | 0       | 0          | 0            | 0                | 6            | 0         |
| <a href="#">Moi Gómez</a>        | MF,FW | 28  | 33 | 2,453 | 3     | 2       | 3          | 0            | 0                | 5            | 0         |
| <a href="#">Sergio Herrera</a>   | GK    | 29  | 17 | 1,530 | 0     | 0       | 0          | 0            | 0                | 3            | 0         |
| <a href="#">Léo Baptistão</a>    | FW,MF | 29  | 28 | 1,726 | 5     | 1       | 5          | 0            | 0                | 7            | 0         |

A list of all players for three soccer leagues from different countries in Europe during the season 2022-2023 was compiled. Each league represents a population and within each League only the players that participated in at least 16 matches, which represents half of the season, were considered; this distinction was made to avoid players in the Original Data Set that were included but that didn't contribute to the overall performance of the league due to injury or transference to a different league, among other possible causes.

The Summary Statistics for the Soccer League in Belgium is presented below.

The complete summary statistics for the populations of the 3 leagues considered is presented in the appendix.

| <b>BELGIUM</b>   | <b>Age</b> | <b>MP</b> | <b>Min</b> | <b>Gls</b> | <b>Ast</b> | <b>G-PK</b> | <b>PK</b> | <b>PKatt</b> | <b>CrdY</b> | <b>CrdR</b> |
|------------------|------------|-----------|------------|------------|------------|-------------|-----------|--------------|-------------|-------------|
| <b>Mean</b>      | 24.74      | 26.25     | 1845.23    | 2.80       | 1.93       | 2.45        | 0.36      | 0.43         | 3.61        | 0.23        |
| <b>Median</b>    | 24.00      | 27.00     | 1891.00    | 1.00       | 1.00       | 1.00        | 0.00      | 0.00         | 3.00        | 0.00        |
| <b>Mode</b>      | 23.00      | 31.00     | 3060.00    | 0.00       | 0.00       | 0.00        | 0.00      | 0.00         | 3.00        | 0.00        |
| <b>Std. Dev.</b> | 4.21       | 5.06      | 660.54     | 3.61       | 2.34       | 3.01        | 1.08      | 1.20         | 2.50        | 0.47        |
| <b>Variance</b>  | 17.75      | 25.61     | 436318.28  | 13.06      | 5.45       | 9.06        | 1.16      | 1.43         | 6.24        | 0.22        |
| <b>Range</b>     | 20.00      | 17.00     | 2781.00    | 20.00      | 21.00      | 16.00       | 6.00      | 6.00         | 12.00       | 2.00        |
| <b>Minimum</b>   | 17.00      | 17.00     | 279.00     | 0.00       | 0.00       | 0.00        | 0.00      | 0.00         | 0.00        | 0.00        |
| <b>Maximum</b>   | 37.00      | 34.00     | 3060.00    | 20.00      | 21.00      | 16.00       | 6.00      | 6.00         | 12.00       | 2.00        |
| <b>Sum</b>       | 6753.00    | 7165.00   | 503749.00  | 765.00     | 527.00     | 668.00      | 97.00     | 118.00       | 985.00      | 62.00       |
| <b>Count</b>     | 273.00     | 273.00    | 273.00     | 273.00     | 273.00     | 273.00      | 273.00    | 273.00       | 273.00      | 273.00      |

## **SAMPLING METHOD**

Five soccer leagues from Europe were selected as populations. For each of the leagues a sample of size 20 was taken, using the simple random sample process. For the randomization of the data a random number generator was used to assign a random number to each of the observations (players). By filtering the data, the players were

arranged by decreasing order of the random number assigned and the first 20 observations were chosen as the sample group. This process was repeated for each of the leagues.

The selection of 20 as size for the sampling is arbitrary and will be compared with a second sampling where the process of selection will be the same, but where the size of the sample will be such that it represents roughly 10% of the population of the respective league. The five leagues considered as well as their population (N), size of the first sample (n1) and size of the second sample (n2) is as follows:

|                | <b>N</b> | <b>n1</b> | <b>n2</b> |
|----------------|----------|-----------|-----------|
| <b>BELGIUM</b> | 273      | 20        | 28        |
| <b>ENGLAND</b> | 346      | 20        | 35        |
| <b>GERMANY</b> | 515      | 20        | 52        |

Simple Random Sampling was used to obtain a sample that represents the overall characteristics of the population. There are categories within the population, like position played, that might be used to do stratified sampling however Simple Random Sampling was chosen because the main objective is to understand the structure of the dataset and overall metrics of the leagues.

## **SAMPLES**

Due to space available and convenience, two variables are included in the summary statistics of both samples ( $n1 = 20$ ,  $n2 = 0.1 * N$ ) as well as the populations

(Leagues). Summary Statistics for all the Variables will be provided in the appendix. The variables chosen are Age of the Players and Goals. Age allows us to understand a characteristic of the league while Goals is an overall measure of performance.

|            | BELGIUM (AGE) |        |        | BELGIUM (GOALS) |        |        |
|------------|---------------|--------|--------|-----------------|--------|--------|
|            | POP           | SMPL 1 | SMPL 2 | POP             | SMPL 1 | SMPL 2 |
| Mean       | 24.74         | 23.25  | 25.04  | 2.80            | 3.00   | 2.93   |
| Std. Error |               | 0.91   | 0.72   |                 | 0.86   | 0.52   |
| Median     | 24.00         | 22.50  | 24.00  | 1.00            | 2.00   | 2.00   |
| Mode       | 23.00         | 25.00  | 23.00  | 0.00            | 0.00   | 1.00   |
| Std. Dev.  | 4.21          | 4.08   | 3.82   | 3.61            | 3.84   | 2.75   |
| Variance   | 17.75         | 16.62  | 14.63  | 13.06           | 14.74  | 7.55   |
| Range      | 20.00         | 17.00  | 15.00  | 20.00           | 16.00  | 9.00   |
| Minimum    | 17.00         | 17.00  | 18.00  | 0.00            | 0.00   | 0.00   |
| Maximum    | 37.00         | 34.00  | 33.00  | 20.00           | 16.00  | 9.00   |
| Sum        | 6753.00       | 465.00 | 701.00 | 765.00          | 60.00  | 82.00  |
| Count      | 273.00        | 20.00  | 28.00  | 273.00          | 20.00  | 28.00  |

|            | ENGLAND (AGE) |        |        | ENGLAND (GOALS) |        |        |
|------------|---------------|--------|--------|-----------------|--------|--------|
|            | POP           | SMPL 1 | SMPL 2 | POP             | SMPL 1 | SMPL 2 |
| Mean       | 25.87         | 24.45  | 25.80  | 2.80            | 1.45   | 2.63   |
| Std. Error |               | 0.93   | 0.56   |                 | 0.44   | 0.80   |
| Median     | 25.00         | 23.50  | 26.00  | 1.00            | 0.50   | 1.00   |
| Mode       | 25.00         | 23.00  | 24.00  | 0.00            | 0.00   | 0.00   |
| Std. Dev   | 3.84          | 4.16   | 3.34   | 4.24            | 1.96   | 4.72   |
| Variance   | 14.75         | 17.31  | 11.16  | 18.02           | 3.84   | 22.24  |
| Range      | 20.00         | 18.00  | 14.00  | 36.00           | 7.00   | 20.00  |
| Minimum    | 17.00         | 19.00  | 18.00  | 0.00            | 0.00   | 0.00   |
| Maximum    | 37.00         | 37.00  | 32.00  | 36.00           | 7.00   | 20.00  |
| Sum        | 8952.00       | 489.00 | 903.00 | 970.00          | 29.00  | 92.00  |
| Count      | 346.00        | 20.00  | 35.00  | 346.00          | 20.00  | 35.00  |

|            | GERMANY (AGE) |        |         | GERMANY (GOALS) |        |        |
|------------|---------------|--------|---------|-----------------|--------|--------|
|            | POP           | SMPL 1 | SMPL 2  | POP             | SMPL 1 | SMPL 2 |
| Mean       | 25.05         | 26.50  | 24.35   | 1.84            | 2.10   | 1.85   |
| Std. Error |               | 1.02   | 0.55    |                 | 0.58   | 0.37   |
| Median     | 25.00         | 26.00  | 25.00   | 1.00            | 1.00   | 1.00   |
| Mode       | 24.00         | 24.00  | 29.00   | 0.00            | 0.00   | 0.00   |
| Std. Dev   | 4.44          | 4.58   | 4.00    | 2.94            | 2.57   | 2.64   |
| Variance   | 19.69         | 21.00  | 16.00   | 8.64            | 6.62   | 7.00   |
| Range      | 22.00         | 19.00  | 16.00   | 16.00           | 8.00   | 11.00  |
| Minimum    | 16.00         | 16.00  | 16.00   | 0.00            | 0.00   | 0.00   |
| Maximum    | 38.00         | 35.00  | 32.00   | 16.00           | 8.00   | 11.00  |
| Sum        | 12902.00      | 530.00 | 1266.00 | 948.00          | 42.00  | 96.00  |
| Count      | 515.00        | 20.00  | 52.00   | 515.00          | 20.00  | 52.00  |

To allow for an easier comparison, for each of the leagues and the respective variables a table with the confidence interval by sample is presented below:

|                 |                                  |              |
|-----------------|----------------------------------|--------------|
| BELGIUM         | <b>TRUE MEAN = 24.74</b>         |              |
| AGE             | <b>CONFIDENCE INTERVAL (95%)</b> | <b>RANGE</b> |
| <b>SAMPLE 1</b> | (21.47 , 25.03)                  | 3.57         |
| <b>SAMPLE 2</b> | (23.63 , 26.65)                  | 2.82         |

|                 |                                  |              |
|-----------------|----------------------------------|--------------|
| BELGIUM         | <b>TRUE MEAN = 2.8</b>           |              |
| GOALS           | <b>CONFIDENCE INTERVAL (95%)</b> | <b>RANGE</b> |
| <b>SAMPLE 1</b> | (1.31 , 4.69)                    | 3.37         |
| <b>SAMPLE 2</b> | (1.91 , 3.95)                    | 2.04         |

|                 |                                  |              |
|-----------------|----------------------------------|--------------|
| ENGLAND         | <b>TRUE MEAN = 25.87</b>         |              |
| AGE             | <b>CONFIDENCE INTERVAL (95%)</b> | <b>RANGE</b> |
| <b>SAMPLE 1</b> | (22.63 , 26.27)                  | 3.65         |
| <b>SAMPLE 2</b> | (24.72 , 26.92)                  | 2.20         |

|         |                        |  |
|---------|------------------------|--|
| ENGLAND | <b>TRUE MEAN = 2.8</b> |  |
|---------|------------------------|--|



| GOALS           | CONFIDENCE INTERVAL (95%) | RANGE |
|-----------------|---------------------------|-------|
| <b>SAMPLE 1</b> | (0.59 , 2.31)             | 1.72  |
| <b>SAMPLE 2</b> | (1.06 , 4.20)             | 3.14  |

| GERMANY         | <b>TRUE MEAN = 25.05</b>  |       |
|-----------------|---------------------------|-------|
| AGE             | CONFIDENCE INTERVAL (95%) | RANGE |
| <b>SAMPLE 1</b> | (24.50 , 28.50)           | 4.00  |
| <b>SAMPLE 2</b> | (23.27 , 25.43)           | 2.16  |

| GERMANY         | <b>TRUE MEAN = 1.84</b>   |       |
|-----------------|---------------------------|-------|
| GOALS           | CONFIDENCE INTERVAL (95%) | RANGE |
| <b>SAMPLE 1</b> | (0.96 , 3.24)             | 2.27  |
| <b>SAMPLE 2</b> | (1.12 , 2.58)             | 1.45  |

## RESULTS

The 3 leagues as populations allow us to compare the effect that a population size has on the confidence interval of a sample. Sample 1, for all three leagues, consists of 20 observations regardless of the population size. Sample 2 has a size that represents the 10% of the respective population size.

|                | <b>N</b> | <b>n1</b> | <b>n2</b> |
|----------------|----------|-----------|-----------|
| <b>BELGIUM</b> | 273      | 20        | 28        |
| <b>ENGLAND</b> | 346      | 20        | 35        |
| <b>GERMANY</b> | 515      | 20        | 52        |

- A first result comes by evaluating if the true mean (mean of the population) is captured by the range of the confidence interval of the respective samples. For the 3 populations and the 2 types of samples there is only one instance where the true mean fails to fall between the confidence interval: For the variable Goals and the Population England, the first sample with 20

observations has a confidence interval of (0.59 , 2.31). The mean of the population is 2.8. This can be explained by understanding how the sample fails to capture the characteristics of the population, as seen by the range of values for Goals of the sample 1 in contrast with the population: 7 v. 32. Thus, the random sample 1 achieved a low variance by selecting observations with values close to each other but that do not represent the population.

- A second result comes from evaluating the range of the respective Confidence Intervals. Between populations, a sample of strictly 20 observations produces a confidence interval with a range that increases when the population size increases in contrast with a sample size that is dependent on the size of the population.

1. Roger Horberry; April 6, 2023; Blog GWI;  
<https://blog.gwi.com/chart-of-the-day/worlds-most-popular-sports/>
2. [https://fbref.com/en/?utm\\_source=fb&utm\\_medium=sr\\_xsite&utm\\_campaign=2023\\_01\\_srnav](https://fbref.com/en/?utm_source=fb&utm_medium=sr_xsite&utm_campaign=2023_01_srnav)

All data computations were obtained using Excel.

All data sets were gathered from FBREF.COM.

## APPENDIX

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